1. hard process (ME): $q\bar{q} \rightarrow b\bar{b}$

run 1 lumi 1 event 1

2 particles drawn (marker size ~ log p_T; label = p_T [GeV] for p_T > 5 GeV)

b quark (status 23): p_T = 113.22 GeV, η = -1.44, ϕ = -2.15

$\bar{b}$ quark (status 23): p_T = 113.22 GeV, η = -3.75, ϕ = 0.99

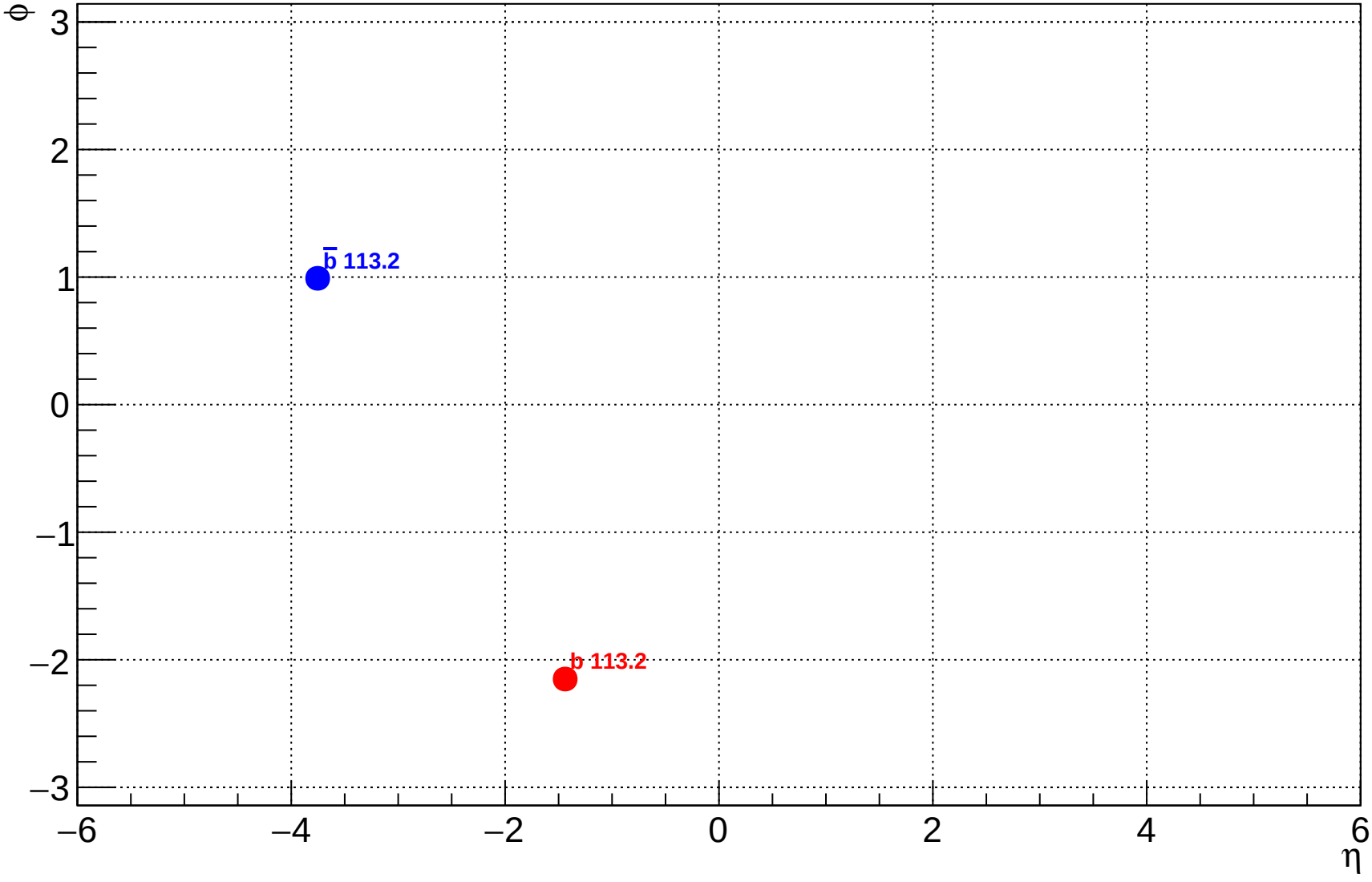
net b+$\bar{b}$: $|\vec{p}_T(b) + \vec{p}_T(\bar{b})|$ = 0.00 GeV (p_x = -0.00, p_y = 0.00), $\Sigma|p_T|$ = 226.44 GeV, $\Delta\phi(b,\bar{b})$ = 3.14

sample ME_PS_MPI_Had_Decay_GenJ

ISR on, FSR on, MPI on

Had on, Decay on

pTHatMin 100 GeV, jet pT > 3 GeV



particle origin (number drawn)

- b system (1)
- $\bar{b}$ system (1)
- b + $\bar{b}$ (vector sum)

marker size:

- p_T = 1 GeV
- p_T = 10 GeV
- p_T = 100 GeV

stages:

1 hard process

2 ISR + FSR

3 MPI / remnants

4 hadronisation

5 hadron decays

2. + parton shower (ISR + FSR)

run 1 lumi 1 event 1

57 particles drawn (marker size ~ log p_T; label = p_T [GeV] for p_T > 5 GeV), 2 with |η| > 6 drawn on the border (x)

b quark (status 52): p_T = 91.59 GeV, η = -1.57, φ = -2.08

$\bar{b}$ quark (status 52): p_T = 118.57 GeV, η = -3.73, φ = 1.02

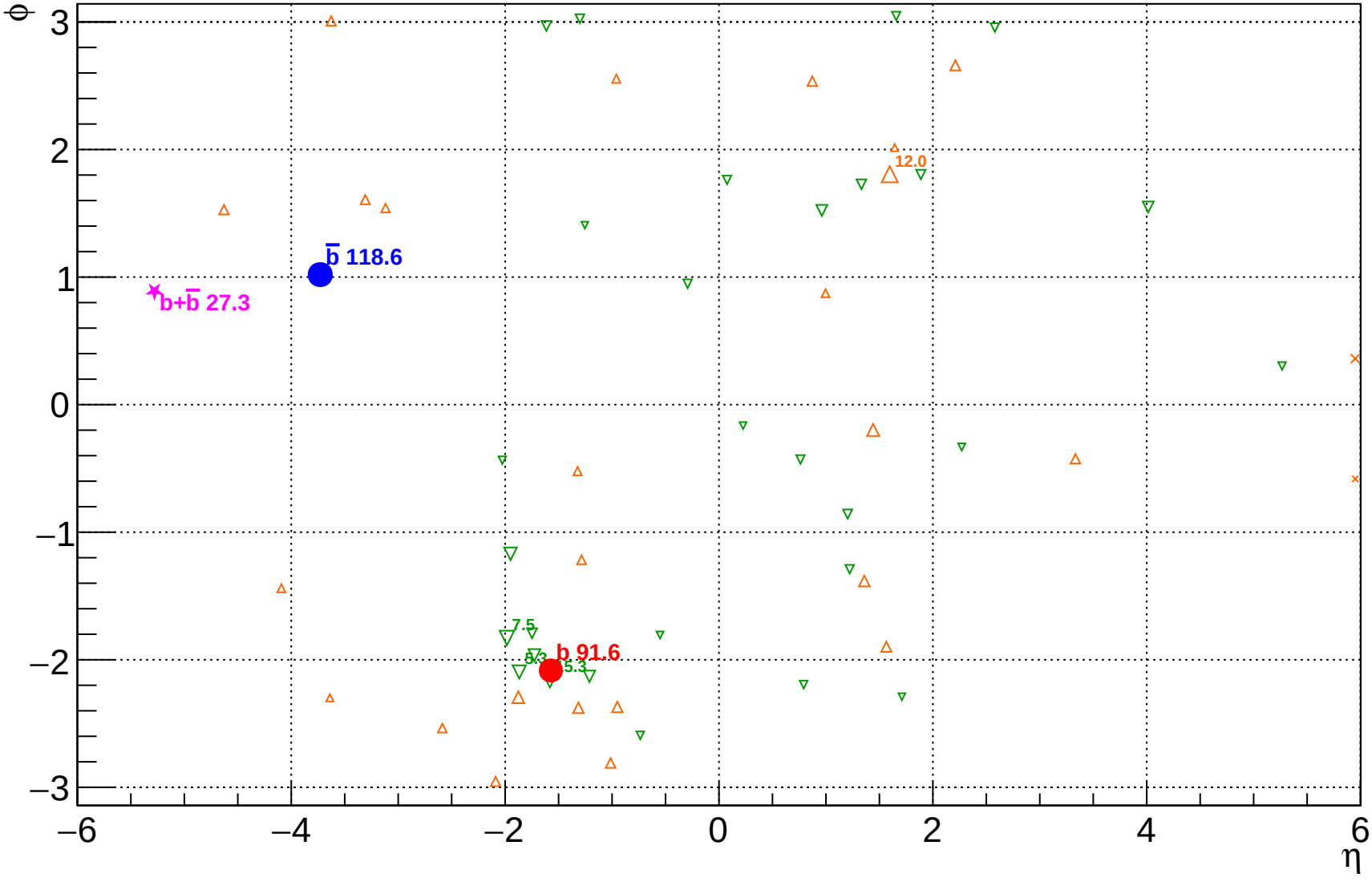
net b+$\bar{b}$: $|\vec{p}_T(b) + \vec{p}_T(\bar{b})|$ = 27.29 GeV (p_x = 17.19, p_y = 21.19), $\Sigma|p_T|$ = 210.16 GeV, $\Delta\phi(b,\bar{b})$ = 3.10

sample ME_PS_MPI_Had_Decay_GenJ

ISR on, FSR on, MPI on

Had on, Decay on

pTHatMin 100 GeV, jet pT > 3 GeV



particle origin (number drawn)

- b system (1)
- $\bar{b}$ system (1)
- b + $\bar{b}$ (vector sum)
- ISR (26)
- FSR (29)

marker size:

- p_T = 1 GeV
- p_T = 10 GeV
- p_T = 100 GeV

stages:

1 hard process

2 ISR + FSR

3 MPI / remnants

4 hadronisation

5 hadron decays

3. + MPI / beam remnants

run 1 lumi 1 event 1

169 particles drawn (marker size ~ log p_T; label = p_T [GeV] for p_T > 5 GeV), 16 with |η| > 6 drawn on the border (x)

b quark (status 62): p_T = 91.74 GeV, η = -1.57, φ = -2.08

$\bar{b}$ quark (status 62): p_T = 116.36 GeV, η = -3.75, φ = 1.01

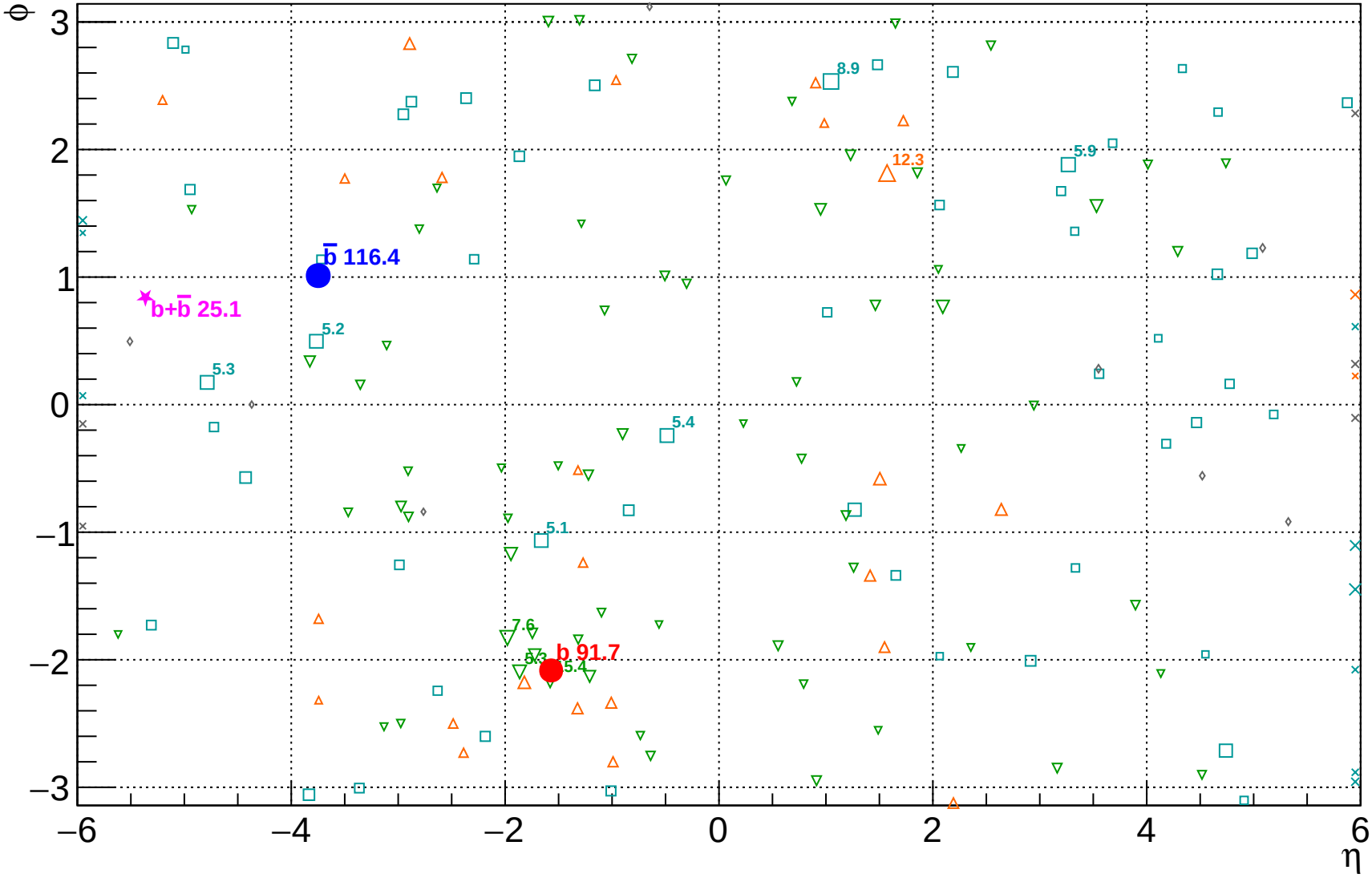
net b+$\bar{b}$: $|\vec{p}_T(b) + \vec{p}_T(\bar{b})|$ = 25.08 GeV (p_x = 16.73, p_y = 18.68), $\Sigma|p_T|$ = 208.10 GeV, $\Delta\phi(b,\bar{b})$ = 3.10

sample ME_PS_MPI_Had_Decay_GenJ

ISR on, FSR on, MPI on

Had on, Decay on

pTHatMin 100 GeV, jet pT > 3 GeV



particle origin (number drawn)

- b system (1)
- $\bar{b}$ system (1)
- b + $\bar{b}$ (vector sum)
- ISR (26)
- FSR (67)
- MPI (61)
- beam remnant (13)

marker size:

- p_T = 1 GeV
- p_T = 10 GeV
- p_T = 100 GeV

stages:

1 hard process

2 ISR + FSR

3 MPI / remnants

4 hadronisation

5 hadron decays

4. + hadronisation (primary hadrons)

run 1 lumi 1 event 1

320 particles drawn (marker size ~ log p_T; label = p_T [GeV] for p_T > 5 GeV), 25 with |η| > 6 drawn on the border (x)

b hadron B^-: p_T = 94.28 GeV, η = -1.57, φ = -2.09

$\bar{b}$ hadron B^{*+}: p_T = 114.17 GeV, η = -3.75, φ = 1.01

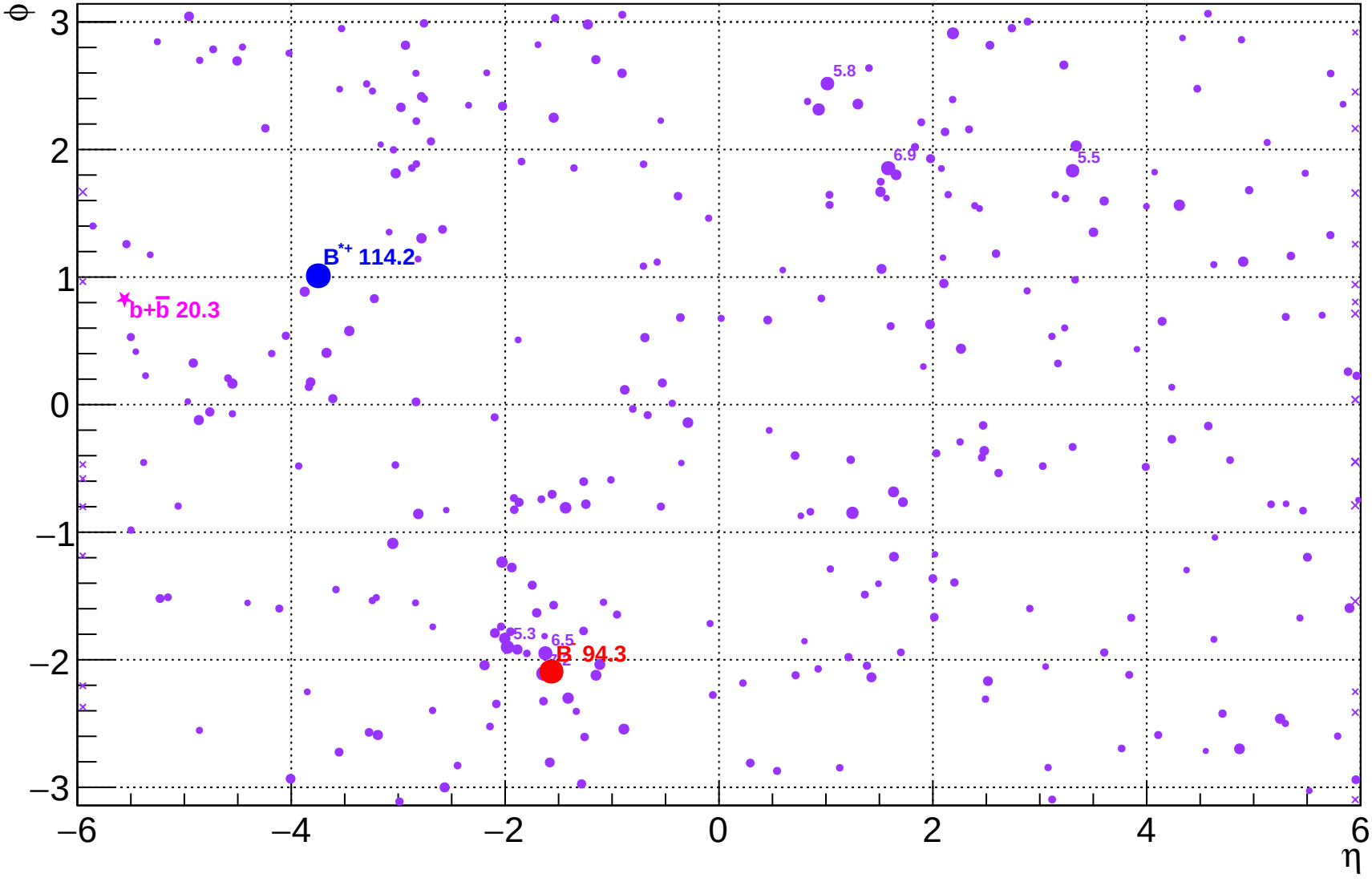
net b+$\bar{b}$: $|\vec{p}_T(b) + \vec{p}_T(\bar{b})|$ = 20.30 GeV (p_x = 13.73, p_y = 14.96), $\Sigma|p_T|$ = 208.45 GeV, $\Delta\phi(b,\bar{b})$ = 3.10

sample ME_PS_MPI_Had_Decay_GenJ

ISR on, FSR on, MPI on

Had on, Decay on

pTHatMin 100 GeV, jet pT > 3 GeV



particle origin (number drawn)

- b system (1)
- $\bar{b}$ system (1)
- b + $\bar{b}$ (vector sum)
- primary hadron (318)

marker size:

- p_T = 1 GeV
- p_T = 10 GeV
- p_T = 100 GeV

stages:

1 hard process

2 ISR + FSR

3 MPI / remnants

4 hadronisation

5 hadron decays

5. + hadron decays (= genParticles status 1)

run 1 lumi 1 event 1

634 particles drawn (marker size ~ log p_T; label = p_T [GeV] for p_T > 5 GeV), 38 with |η| > 6 drawn on the border (x)

b hadron B^- decay products (n=9): p_T = 94.28 GeV, η = -1.57, φ = -2.09

$\bar{b}$ hadron B^{*+} decay products (n=16): p_T = 114.17 GeV, η = -3.75, φ = 1.01

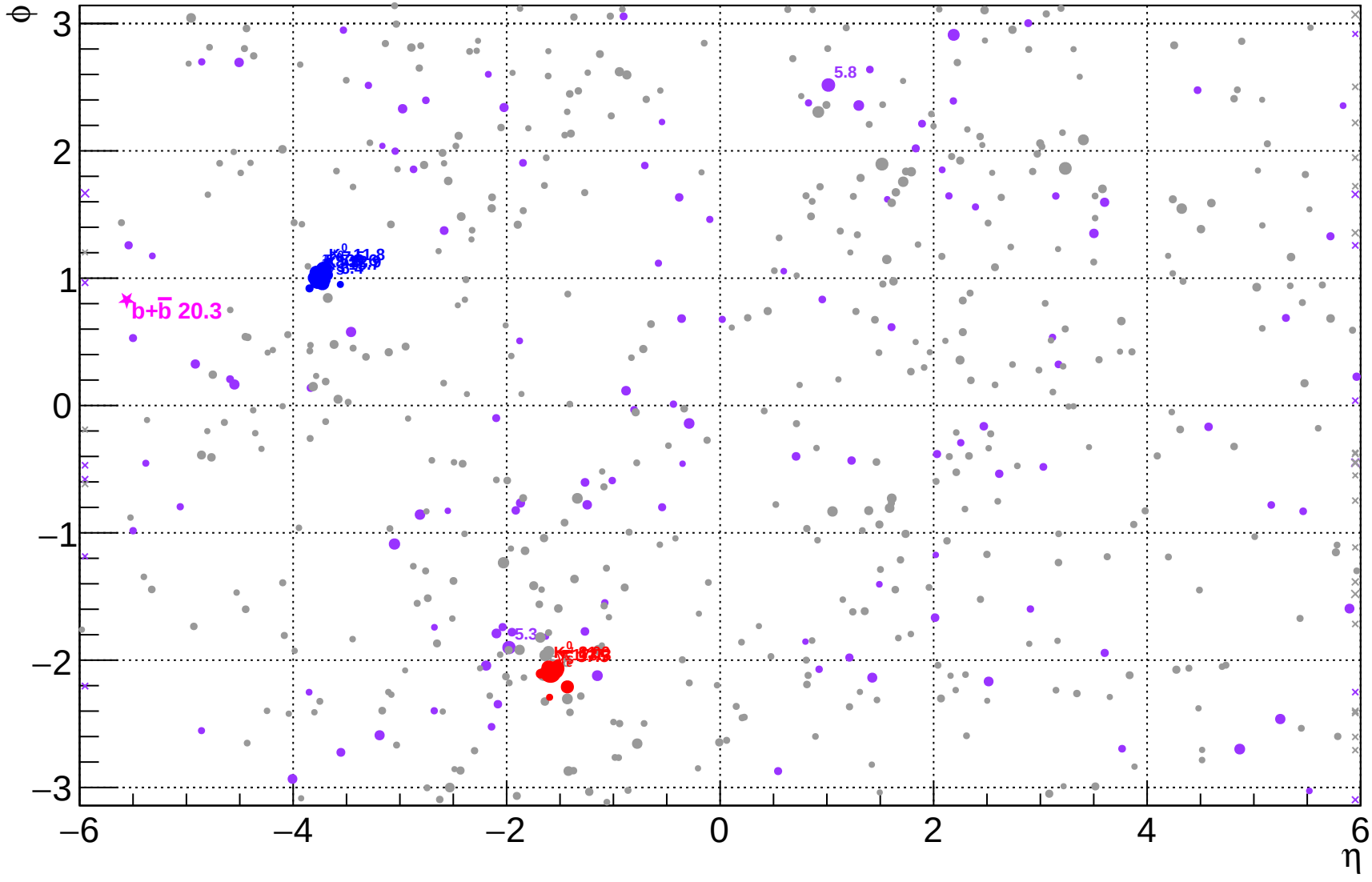
net b+$\bar{b}$: $|\vec{p}_T(b) + \vec{p}_T(\bar{b})|$ = 20.30 GeV (p_x = 13.73, p_y = 14.96), $\Sigma|p_T|$ = 208.45 GeV, $\Delta\phi(b,\bar{b})$ = 3.10

sample ME_PS_MPI_Had_Decay_GenJ

ISR on, FSR on, MPI on

Had on, Decay on

pTHatMin 100 GeV, jet pT > 3 GeV



particle origin (number drawn)

- b system (9)
- $\bar{b}$ system (16)
- b + $\bar{b}$ (vector sum)
- primary hadron (132)
- decay product (477)

marker size:

- p_T = 1 GeV
- p_T = 10 GeV
- p_T = 100 GeV

stages:

1 hard process

2 ISR + FSR

3 MPI / remnants

4 hadronisation

5 hadron decays

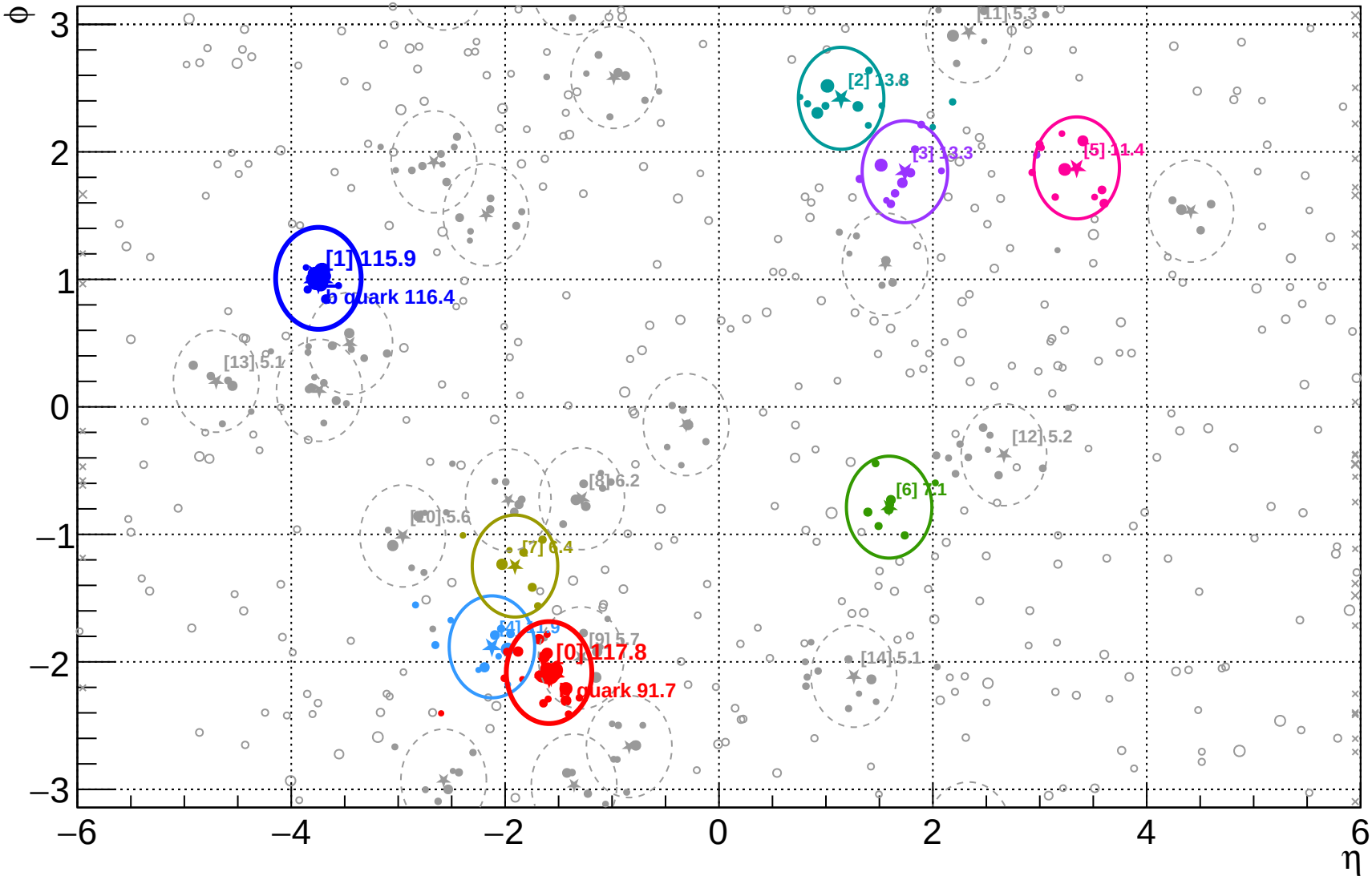
ak4GenJets (process GENJET): anti-k$_T$ R = 0.4, jet p$_T$ > 3 GeV, input genParticlesForJets

27 jets; 239 of 634 status 1 particles clustered; marker size ~ log p$_T$; label = [jet] p$_T$ [GeV] (p$_T$ > 5 GeV); 38 beyond |η| = 6 on the border (x)

b quark (status 71): p$_T$ = 91.74 GeV, η = -1.57, φ = -2.08 → jet [0]: p$_T$ = 117.79 GeV, η = -1.59, φ = -2.08, n = 24, ΔR = 0.019, p$_T$(jet) / p$_T$(q) = 1.284

$\bar{b}$ quark (status 71): p$_T$ = 116.36 GeV, η = -3.75, φ = 1.01 → jet [1]: p$_T$ = 115.94 GeV, η = -3.75, φ = 1.01, n = 18, ΔR = 0.004, p$_T$(jet) / p$_T$(q) = 0.996

b jet + $\bar{b}$ jet: $|\vec{p}_T(b) + \vec{p}_T(\bar{b})|$ = 6.06 GeV (p$_x$ = 4.01, p$_y$ = -4.54), Σ|p$_T$| = 233.73 GeV, Δφ = 3.09, m = 386.8 GeV



run 1 lumi 1 event 1

sample ME_PS_MPI_Had_Decay_GenJe

ISR on, FSR on, MPI on

Had on, Decay on

pTHatMin 100 GeV, jet pT > 3 GeV

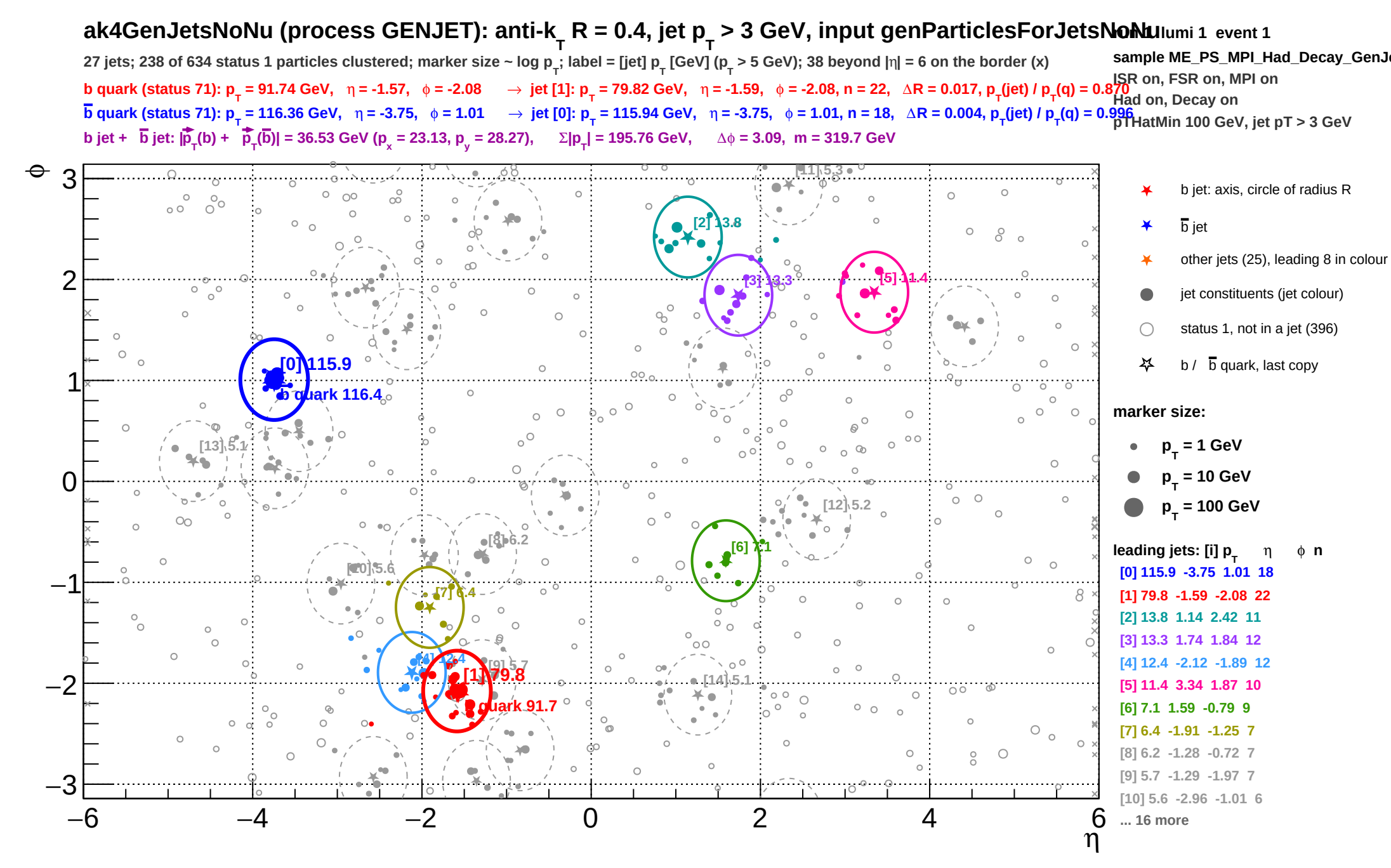
- b jet: axis, circle of radius R
- $\bar{b}$ jet
- other jets (25), leading 8 in colour
- jet constituents (jet colour)
- status 1, not in a jet (395)
- b / $\bar{b}$ quark, last copy

marker size:

- p$_T$ = 1 GeV
- p$_T$ = 10 GeV
- p$_T$ = 100 GeV

leading jets: [i] p$_T$ η φ n

[0] 117.8 -1.59 -2.08 24
[1] 115.9 -3.75 1.01 18
[2] 13.8 1.14 2.42 11
[3] 13.3 1.74 1.84 12
[4] 11.9 -2.12 -1.88 11
[5] 11.4 3.34 1.87 10
[6] 7.1 1.59 -0.79 9
[7] 6.4 -1.91 -1.25 7
[8] 6.2 -1.28 -0.72 7
[9] 5.7 -1.29 -1.97 7
[10] 5.6 -2.96 -1.01 6
... 16 more


ak4GenJetsNoNu (process GENJET): anti-k_T R = 0.4, jet p_T > 3 GeV, input genParticlesForJetsNoNu
27 jets; 238 of 634 status 1 particles clustered; marker size ~ log p_T; label = [jet] p_T [GeV] (p_T > 5 GeV); 38 beyond |η| = 6 on the border (x)
b quark (status 71): p_T = 91.74 GeV, η = -1.57, φ = -2.08 → jet [1]: p_T = 79.82 GeV, η = -1.59, φ = -2.08, n = 22, ΔR = 0.017, p_T(jet) / p_T(q) = 0.870
b̄ quark (status 71): p_T = 116.36 GeV, η = -3.75, φ = 1.01 → jet [0]: p_T = 115.94 GeV, η = -3.75, φ = 1.01, n = 18, ΔR = 0.004, p_T(jet) / p_T(q) = 0.996
b jet + b̄ jet: |p_T(b) + p_T(b̄)| = 36.53 GeV (p_x = 23.13, p_y = 28.27), Σ|p_T| = 195.76 GeV, Δφ = 3.09, m = 319.7 GeV
MPI Lumi 1 event 1
sample ME_PS_MPI_Had_Decay_GenJe
ISR on, FSR on, MPI on
Had on, Decay on
pTHatMin 100 GeV, jet pT > 3 GeV
[0] 115.9
b̄ quark 116.4
[1] 79.8
b quark 91.7
[2] 13.8
[3] 13.3
[5] 11.4
[6] 7.1
[7] 6.4
[8] 6.2
[9] 5.7
[10] 5.6
[11] 5.3
[12] 5.2
[13] 5.1
[14] 5.1
b jet: axis, circle of radius R
b̄ jet
other jets (25), leading 8 in colour
jet constituents (jet colour)
status 1, not in a jet (396)
b / b̄ quark, last copy
marker size:
p_T = 1 GeV
p_T = 10 GeV
p_T = 100 GeV
leading jets: [i] p_T η φ n
[0] 115.9 -3.75 1.01 18
[1] 79.8 -1.59 -2.08 22
[2] 13.8 1.14 2.42 11
[3] 13.3 1.74 1.84 12
[4] 12.4 -2.12 -1.89 12
[5] 11.4 3.34 1.87 10
[6] 7.1 1.59 -0.79 9
[7] 6.4 -1.91 -1.25 7
[8] 6.2 -1.28 -0.72 7
[9] 5.7 -1.29 -1.97 7
[10] 5.6 -2.96 -1.01 6
... 16 more
φ
η
3
2
1
0
−1
−2
−3
−6
−4
−2
0
2
4
6

ak8GenJets (process GENJET): anti-k$_T$ R = 0.8, jet p$_T$ > 3 GeV, input genParticlesForJets

28 jets; 488 of 634 status 1 particles clustered; marker size ~ log p$_T$; label = [jet] p$_T$ [GeV] (p$_T$ > 5 GeV); 39 beyond |η| = 6 on the border (x)

b quark (status 71): p$_T$ = 91.74 GeV, η = -1.57, φ = -2.08 → jet [0]: p$_T$ = 141.10 GeV, η = -1.65, φ = -2.05, n = 55, ΔR = 0.088, p$_T$(jet) / p$_T$(q) = 1.538

$\bar{b}$ quark (status 71): p$_T$ = 116.36 GeV, η = -3.75, φ = 1.01 → jet [1]: p$_T$ = 121.01 GeV, η = -3.75, φ = 0.99, n = 32, ΔR = 0.022, p$_T$(jet) / p$_T$(q) = 1.040

b jet + $\bar{b}$ jet: $|\vec{p}_T(b) + \vec{p}_T(\bar{b})|$ = 24.03 GeV (p$_x$ = 1.27, p$_y$ = -24.00), Σ|p$_T$| = 262.11 GeV, Δφ = 3.04, m = 439.5 GeV

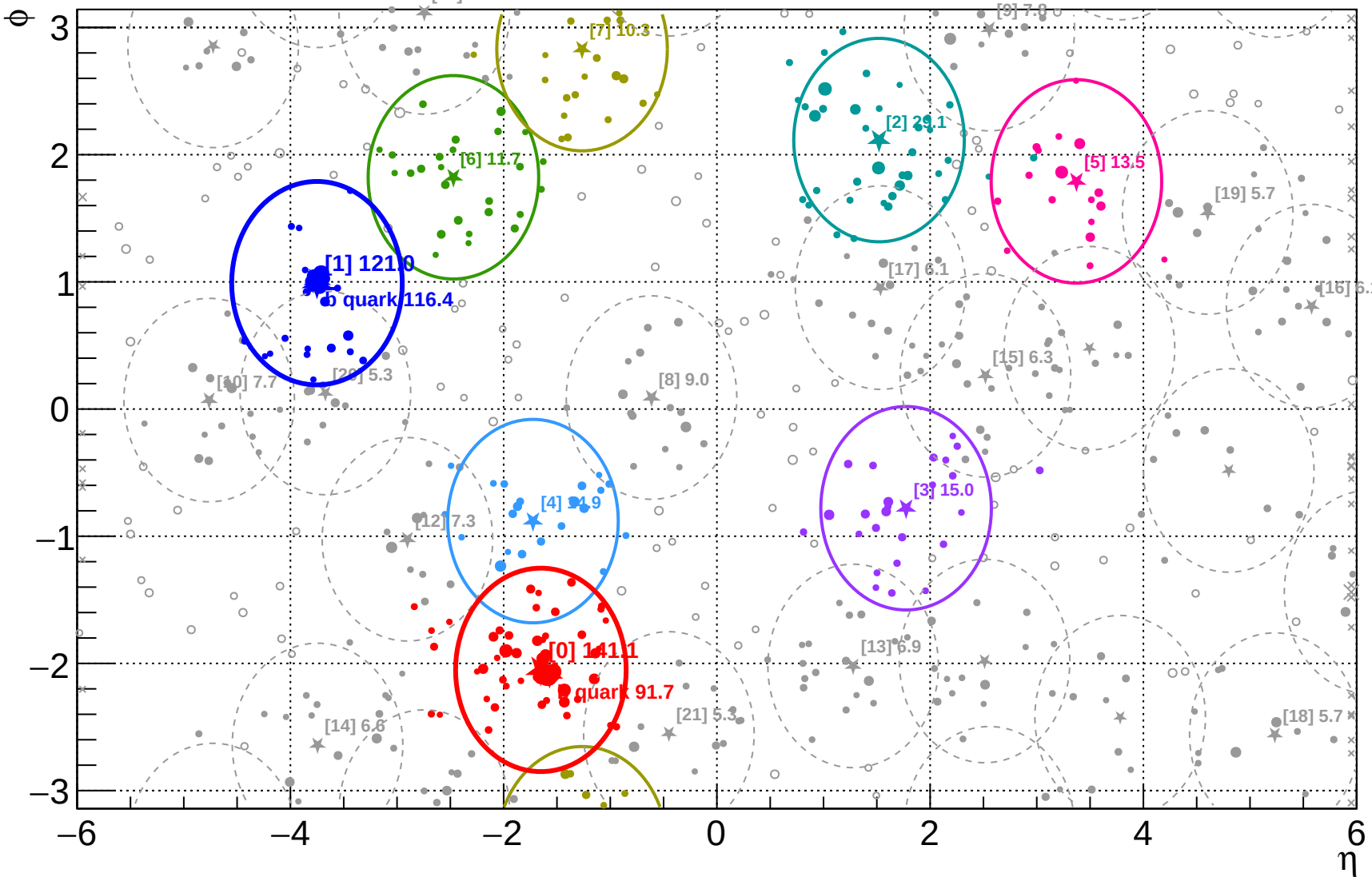
run 1 lumi 1 event 1

sample ME_PS_MPI_Had_Decay_GenJe

ISR on, FSR on, MPI on

Had on, Decay on

pTHatMin 100 GeV, jet pT > 3 GeV



b jet: axis, circle of radius R

$\bar{b}$ jet

other jets (26), leading 8 in colour

jet constituents (jet colour)

status 1, not in a jet (146)

b / $\bar{b}$ quark, last copy

marker size:

p$_T$ = 1 GeV

p$_T$ = 10 GeV

p$_T$ = 100 GeV

leading jets: [i] p$_T$ η φ n

[0] 141.1 -1.65 -2.05 55

[1] 121.0 -3.75 0.99 32

[2] 29.1 1.52 2.11 37

[3] 15.0 1.77 -0.78 26

[4] 14.9 -1.73 -0.88 21

[5] 13.5 3.37 1.79 17

[6] 11.7 -2.47 1.82 26

[7] 10.3 -1.26 2.83 24

[8] 9.0 -0.61 0.09 15

[9] 7.8 2.55 2.99 11

[10] 7.7 -4.76 0.07 14

... 17 more